

Further Mechanics 1

Notes · Edexcel A Level Further Mathematics — Further Mechanics 1 · Chapter 2: Work, Energy and Power

Chapter 2: Work, Energy and Power

Why this chapter matters

Why this matters. Momentum conservation (Chapter 1) tracks a vector quantity through a collision or impulse; work and energy give you a **scalar** toolkit that is often far quicker for problems involving speed changes over a distance, especially once variable forces or gravity are involved. Energy methods will reappear constantly in Chapters 3–5, so fluency here pays off across the whole unit.

Key vocabulary

Term	Meaning
Work done by a force	The energy transferred when a force moves its point of application through a displacement
Kinetic energy (KE)	Energy a body possesses due to its motion
Gravitational potential energy (GPE)	Energy a body possesses due to its position in a gravitational field, relative to some reference level
Work–energy principle	The total work done on a particle equals its change in kinetic energy
Power	The rate at which work is done, or the rate of transfer of energy

Key formula table

Quantity	Formula & notes
Work done by constant force	$W = Fs \cos\theta$ — θ = angle between force and displacement; F , s in N, m; W in joules
Work done against gravity	$W = mgh$ — h = vertical rise
Kinetic energy	$KE = \frac{1}{2}mv^2$ — v = speed
Gravitational potential energy	$GPE = mgh$ — h measured from a chosen reference level
Work–energy principle	Work done by all forces = $\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$ — equivalent to conservation of mechanical energy when only gravity/normal reaction/tension do work and no resistance acts

Power (constant force, constant velocity)	$P = Fv$ — F = driving force, v = speed
Power (general)	$P = dW/dt$ — instantaneous rate of doing work

Concept: Work done by a force

Work is done only by the **component of a force along the direction of motion**. For a force F acting at angle θ to the displacement s :

$$W = Fs \cos\theta$$

A force perpendicular to the motion (e.g. the normal reaction on a particle moving along a horizontal surface, or the tension in a string during circular motion) does **zero** work. A force acting against the direction of motion (e.g. friction, resistance) does **negative** work on the particle, or equivalently the particle does positive work against it.

Worked example 1 — work done at an angle, and the work–energy principle
EXAMPLE 2.1

Question. A crate of mass 20 kg is pulled 15 m across a rough horizontal floor by a force of 60 N acting at 30° above the horizontal. The coefficient of friction between the crate and the floor is 0.3. Given the crate starts from rest, find its speed after 15 m. (Take $g = 9.8 \text{ m s}^{-2}$.)

Step 1. Resolve vertically to find the normal reaction R . The applied force has a vertical component $60 \sin 30^\circ = 30 \text{ N}$ upward, so:

$$R + 30 = 20(9.8) = 196 \implies R = 166 \text{ N} \quad \text{M1A1}$$

Step 2. Friction force $F = \mu R = 0.3 \times 166 = 49.8 \text{ N}$, acting to oppose motion. **M1A1**

Step 3. Find the work done by each force over 15 m:

- Applied force: $W_1 = 60 \times 15 \times \cos 30^\circ = 779.4 \text{ J}$ **M1**
- Friction: $W_2 = -49.8 \times 15 = -747 \text{ J}$ **M1**
- Normal reaction and weight: both perpendicular to motion, so do zero work. **B1**

Step 4. Apply the work–energy principle. Total work done = ΔKE :

$$779.4 - 747 = \frac{1}{2}(20)v^2 - 0 \quad \text{M1}$$

$$32.4 = 10v^2 \implies v^2 = 3.24 \implies v = 1.8 \text{ m s}^{-1} \quad \text{A1}$$

Answer: speed after 15 m is 1.8 m s^{-1} .

EXAM TIP

This is the example that separates genuinely understanding “work” from having memorised $W = Fs$. Write out, force by force, whether each one does positive, negative or zero work *before* plugging in numbers — a quick sketch with the crate, the 30° force, friction arrow, weight and normal reaction drawn on it prevents almost every error here, including the classic mistake of using the full 60 N (rather than its horizontal component) in the work-done calculation, or forgetting to include the vertical component of the pull when finding R .

Concept: Gravitational potential energy and energy conservation

When a particle rises through height h , it gains $GPE = mgh$ (and loses this GPE if it falls). If gravity is the only force doing work (no resistance, no other applied force), **mechanical energy is conserved**:

$$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2$$

Where resistances or other forces also act, use the general form: $(KE + GPE)$ at the end = $(KE + GPE)$ at the start + work done by other forces (with the correct sign).

Worked example 2 — energy conservation on a slope with resistance
EXAMPLE 2.2

Question. A skier of mass 65 kg starts from rest at the top of a slope and skis 40 m down a slope inclined at 20° to the horizontal. A constant resistance of 50 N acts on the skier throughout. Find the skier's speed at the bottom of the slope. (Take $g = 9.8 \text{ m s}^{-2}$.)

Step 1. Vertical height lost: $h = 40 \sin 20^\circ = 13.68 \text{ m}$ (3 s.f.) **M1A1**

Step 2. GPE lost = $mgh = 65 \times 9.8 \times 13.68 = 8714 \dots \text{ J}$ (this converts into KE, minus work done against resistance). **M1**

Step 3. Work done against resistance = $50 \times 40 = 2000 \text{ J}$. **B1**

Step 4. By the work–energy principle: KE gained = GPE lost – work done against resistance:

$$\frac{1}{2}(65)v^2 = 8714 - 2000 = 6714 \text{ (3 s.f.)} \quad \text{M1}$$

$$v^2 = 2(6714)/65 = 206.6 \dots \implies v = 14.4 \text{ m s}^{-1} \text{ (3 s.f.)} \quad \text{A1}$$

Answer: the skier's speed at the bottom is approximately 14.4 m s^{-1} .

EXAM TIP

Energy methods are attractive because they avoid resolving forces along the slope — and they still work perfectly when acceleration is not constant along the path (e.g. a curved slope), where kinematics (SUVAT) would fail. Try imagining the same problem with a curved ski-jump profile: energy methods don't care about the shape of the path, only the start and end heights — this is often the point at which the penny drops.

Concept: Power

For a constant driving force F moving a particle at constant speed v , power $P = Fv$. More generally, if a vehicle has driving force $F(t)$ and speed $v(t)$, the instantaneous power is $P = Fv$ at that instant.

A very common exam scenario: a vehicle's engine works at **constant power** P , so as its speed changes, the driving force must change too ($F = P/v$) to keep P fixed. This is essential for problems involving acceleration up a slope, where the driving force is not constant.

Worked example 3 — constant power, finding driving force and using Newton's second law
EXAMPLE 2.3

Question. A car of mass 900 kg has an engine working at a constant power of 24 kW. The car experiences a constant resistance to motion of 400 N. Find the car's acceleration at the instant its speed is 20 m s^{-1} , and find the car's maximum possible speed on a level road.

Step 1. At speed 20 m s^{-1} , driving force $F = P/v = 24000/20 = 1200 \text{ N}$. **M1A1**

Step 2. Apply Newton's second law along the direction of motion:

$$F - \text{resistance} = ma$$

$$1200 - 400 = 900a \quad \text{M1}$$

$$800 = 900a \implies a = 8/9 \approx 0.889 \text{ m s}^{-2} \quad \text{A1}$$

Step 3. Maximum speed occurs when acceleration = 0, i.e. driving force = resistance = 400 N. Then:

$$P = Fv \implies 24000 = 400v \quad \text{M1}$$

$$v = 60 \text{ m s}^{-1} \quad \text{A1}$$

Answer: acceleration at 20 m s^{-1} is $8/9 \approx 0.89 \text{ m s}^{-2}$; maximum speed is 60 m s^{-1} .

EXAM TIP

“Maximum speed = zero acceleration” is one of the most reliable mark-winning ideas in this topic, but it needs to be understood as a concept, not a formula: at maximum speed the car can no longer accelerate, so the resultant force is zero, so driving force exactly balances resistance. Once you see that argument you can apply it to any “maximum speed” question (with resistance proportional to speed, up a slope, etc.) without needing a separate formula for each variant.

Worked example 4 — constant power on an incline

EXAMPLE 2.4

Question. A car of mass 1200 kg has an engine working at a constant power of 36 kW as it climbs a hill inclined at 5° to the horizontal. The resistance to motion is constant at 500 N . Find the car's maximum speed up the hill. (Take $g = 9.8 \text{ m s}^{-2}$.)

Step 1. At maximum speed the acceleration is zero, so the driving force exactly balances the resistance **and** the component of weight acting down the slope:

$$F = \text{resistance} + mg \sin 5^\circ \quad \text{M1}$$

Step 2. Component of weight down the slope: $mg \sin 5^\circ = 1200 \times 9.8 \times \sin 5^\circ = 1025 \text{ N}$ (3 s.f.)
A1

$$\text{So } F = 500 + 1025 = 1525 \text{ N (3 s.f.)} \quad \text{A1}$$

Step 3. Use $P = Fv$ at maximum speed:

$$36000 = 1525v \quad \text{M1}$$

$$v = 23.6 \text{ m s}^{-1} \text{ (3 s.f.)} \quad \text{A1}$$

Answer: the car's maximum speed up the hill is approximately 23.6 m s^{-1} .

EXAM TIP

This is the natural extension once the flat-road version (Worked example 3) feels secure — and it is a very common exam variation. The only new idea is that “resultant force = 0 at max speed” now has to include the weight component $mg \sin \theta$ alongside resistance, not just resistance

alone. Redraw your force diagram for the incline case before writing any equation: once F , resistance and $mg \sin\theta$ are all marked along the slope, the equation $F = \text{resistance} + mg \sin\theta$ (going uphill) or $F + mg \sin\theta = \text{resistance}$ (going downhill, if driving force still acts up the slope) becomes visual rather than something to memorise.

Chapter summary

Term	Meaning
Work done by a force	$W = Fs \cos\theta$; only the component along the motion contributes
Kinetic energy	$KE = \frac{1}{2}mv^2$
Gravitational potential energy	$GPE = mgh$, relative to a chosen level
Work–energy principle	total work done = change in KE; equivalent to energy conservation when only gravity acts
Power	$P = Fv$ for constant force/speed; $P = dW/dt$ in general; “max speed” $\Leftrightarrow$ zero acceleration $\Leftrightarrow$ driving force = resistance

Common exam mistakes

- Using the full magnitude of a force in $W = Fs$ instead of resolving to find the component along the direction of motion.
- Forgetting that the normal reaction and any force perpendicular to motion do zero work.
- Sign errors when a particle loses height (GPE decreases) or moves against resistance (work done is negative).
- Using $P = Fv$ with a non-constant force/speed without realising F must be recalculated at the instant in question.
- Assuming “maximum speed” means the point where distance is greatest, rather than where acceleration is zero.

You should now be able to

- Calculate work done by a constant force acting at an angle to the direction of motion, including cases where a force does zero or negative work.
- Apply the work–energy principle to find unknown speeds, distances or forces, including problems with friction or resistance.
- Use conservation of mechanical energy for particles moving under gravity alone, and adapt it when resistances also act.
- Use $P = Fv$ to relate power, driving force and speed, and apply the “zero acceleration at maximum speed” argument.